I

Consider N distinguishable, non-interacting particles, each of which can occupy one of two energy levels, $+\epsilon$ or $-\epsilon$, where $\epsilon > 0$. We first use the microcanonical ensemble and then check the result using the canonical ensemble.

(1)

Let N_+ and N_- denote the numbers of particles occupying the $+\epsilon$ and $-\epsilon$ levels, respectively. Conservation of the total number of particles gives

$$\boxed{N = N_+ + N_-}.$$

(2)

The total energy is

$$\boxed{E = \epsilon(N_+ - N_-)}.$$

(3)

Solving the two equations above,

$$N_+ = \frac{1}{2}\left(N + \frac{E}{\epsilon}\right), \qquad N_- = \frac{1}{2}\left(N - \frac{E}{\epsilon}\right).$$

Because the particles are distinguishable, the number of microstates with specified N_+ and N_- is the binomial coefficient

$$\begin{aligned} W(E,N) &= \binom{N}{N_+} = \frac{N!}{N_+!N_-!} \\ &= \boxed{\frac{N!}{\left[\frac{1}{2}\left(N + \frac{E}{\epsilon}\right)\right]!\left[\frac{1}{2}\left(N - \frac{E}{\epsilon}\right)\right]!}}. \end{aligned}$$

The factorial expression is understood for the allowed discrete energies, for which N_+ and N_- are non-negative integers.

(4)

From Boltzmann's principle, $S = k \ln W$. Applying Stirling's formula, $\ln n! \simeq n \ln n - n$, we obtain

$$S(E,N) \simeq k\left[N \ln N - N_+ \ln N_+ - N_- \ln N_-\right].$$

Introduce the fractions

$$p_\pm \equiv \frac{N_\pm}{N} = \frac{1}{2}\left(1 \pm \frac{E}{N\epsilon}\right).$$

Since $p_+ + p_- = 1$, the $\ln N$ terms cancel:

$$S(E,N) = -kN\left(p_+ \ln p_+ + p_- \ln p_-\right).$$

Thus, explicitly,

$$\boxed{S(E,N) = -kN\left[\frac{1}{2}\left(1+\frac{E}{N\epsilon}\right)\ln\left(\frac{1}{2}\left(1+\frac{E}{N\epsilon}\right)\right) + \frac{1}{2}\left(1-\frac{E}{N\epsilon}\right)\ln\left(\frac{1}{2}\left(1-\frac{E}{N\epsilon}\right)\right)\right].}$$

(5)

Because N_+ must be an integer,

$$E = \epsilon(N_+ - N_-) = -N\epsilon + 2N_+\epsilon, \qquad N_+ = 0, 1, \ldots, N.$$

Therefore, successive allowed values of E are separated by

$$\boxed{\Delta E = 2\epsilon}.$$

For large N, we treat E as a continuous variable when taking thermodynamic derivatives.

(6)

The microcanonical definition of temperature is

$$\frac{1}{T} = \left(\frac{\partial S}{\partial E}\right)_N.$$

From the entropy expression,

$$\begin{aligned}\frac{\partial S}{\partial E} &= -kN\left[(\ln p_+ + 1)\frac{\partial p_+}{\partial E} + (\ln p_- + 1)\frac{\partial p_-}{\partial E}\right] \\ &= -\frac{k}{2\epsilon}\left(\ln p_+ - \ln p_-\right),\end{aligned}$$

where $\partial p_\pm/\partial E = \pm(2N\epsilon)^{-1}$. Hence,

$$\boxed{\frac{1}{T} = -\frac{k}{2\epsilon}\ln\left(\frac{N\epsilon + E}{N\epsilon - E}\right)}.$$

(7)

Rearranging the result from (6),

$$\begin{aligned}\ln\left(\frac{N\epsilon + E}{N\epsilon - E}\right) &= -\frac{2\epsilon}{kT}, \\ \frac{N\epsilon + E}{N\epsilon - E} &= e^{-2\epsilon/(kT)}.\end{aligned}$$

Solving for E,

$$\begin{aligned}E &= -N\epsilon\,\frac{1 - e^{-2\epsilon/(kT)}}{1 + e^{-2\epsilon/(kT)}} \\ &= -N\epsilon\tanh\left(\frac{\epsilon}{kT}\right).\end{aligned}$$

Therefore,

$$\boxed{E(N,T) = -N\epsilon\tanh\left(\frac{\epsilon}{kT}\right)}.$$

(8)

The heat capacity at fixed N is

$$\begin{aligned} C &= \left(\frac{\partial E}{\partial T}\right)_N \\ &= -N\epsilon \,\mathrm{sech}^2\!\left(\frac{\epsilon}{kT}\right)\left(-\frac{\epsilon}{kT^2}\right). \end{aligned}$$

Thus,

$$\boxed{C = \frac{N\epsilon^2}{kT^2}\,\mathrm{sech}^2\!\left(\frac{\epsilon}{kT}\right)}.$$

(9)

We now use the canonical ensemble. For a single particle,

$$z_1 = e^{-\beta\epsilon} + e^{\beta\epsilon} = 2\cosh(\beta\epsilon), \qquad \beta \equiv \frac{1}{kT}.$$

Since the particles are distinguishable and independent,

$$\boxed{Z = z_1^N = [2\cosh(\beta\epsilon)]^N}.$$

(10)

The mean energy in the canonical ensemble is

$$\begin{aligned} E &= -\frac{\partial \ln Z}{\partial \beta} \\ &= -N\frac{\partial}{\partial \beta}\ln[2\cosh(\beta\epsilon)] \\ &= -N\epsilon\tanh(\beta\epsilon). \end{aligned}$$

Therefore,

$$\boxed{E = -N\epsilon\tanh\!\left(\frac{\epsilon}{kT}\right)},$$

in agreement with the microcanonical result in (7).

Note: bounded energy and negative temperature. The total energy is bounded, $-N\epsilon \le E \le N\epsilon$. The entropy reaches its maximum near $E = 0$. Hence the microcanonical slope $(\partial S/\partial E)_N$ is positive for $E < 0$ but negative for $E > 0$. The latter corresponds to a negative statistical temperature (population inversion), which is possible for this bounded spectrum. The heat-capacity formula in (8) remains non-negative.

II

Now consider two particles and three non-degenerate single-particle levels with energies 0, ϵ, and 3ϵ. We enumerate the available two-particle microstates for each kind of statistics. Spin is ignored.

(11)

Distinguishable particles. Label the particles A and B. Their ordered energy pairs are

$$(0,0),\ (0,\epsilon),\ (\epsilon,0),\ (0,3\epsilon),\ (3\epsilon,0),\ (\epsilon,\epsilon),$$
$$(\epsilon,3\epsilon),\ (3\epsilon,\epsilon),\ (3\epsilon,3\epsilon).$$

There are nine microstates, including distinct permutations of the energies of A and B. Consequently,

$$\begin{aligned} Z_{\text{dist}} &= 1 + 2e^{-\beta\epsilon} + e^{-2\beta\epsilon} + 2e^{-3\beta\epsilon} + 2e^{-4\beta\epsilon} + e^{-6\beta\epsilon} \\ &= \boxed{\left(1 + e^{-\beta\epsilon} + e^{-3\beta\epsilon}\right)^2}. \end{aligned}$$

(12)

Bose–Einstein statistics. For indistinguishable bosons, a microstate is specified by the occupation numbers $(n_0, n_\epsilon, n_{3\epsilon})$, with $n_0 + n_\epsilon + n_{3\epsilon} = 2$. All six possibilities are

$(n_0, n_\epsilon, n_{3\epsilon})$	$(2,0,0)$	$(1,1,0)$	$(0,2,0)$	$(1,0,1)$	$(0,1,1)$	$(0,0,2)$
E_{tot}/ϵ	0	1	2	3	4	6

Multiple bosons may occupy the same one-particle state. Thus,

$$\boxed{Z_{\text{BE}} = 1 + e^{-\beta\epsilon} + e^{-2\beta\epsilon} + e^{-3\beta\epsilon} + e^{-4\beta\epsilon} + e^{-6\beta\epsilon}}.$$

(13)

Fermi–Dirac statistics. For indistinguishable fermions without spin, Pauli exclusion requires $n_0, n_\epsilon, n_{3\epsilon} \in \{0, 1\}$. The three allowed occupation-number configurations are

$(n_0, n_\epsilon, n_{3\epsilon})$	$(1,1,0)$	$(1,0,1)$	$(0,1,1)$
E_{tot}/ϵ	1	3	4

Hence,

$$\boxed{Z_{\text{FD}} = e^{-\beta\epsilon} + e^{-3\beta\epsilon} + e^{-4\beta\epsilon}}.$$

Note: how microstates are counted. For distinguishable particles, swapping the two particles produces a new microstate. For identical bosons or fermions, permutations do not produce new states: we count occupation-number configurations instead. Bosons allow occupations $0, 1, 2, \ldots$; spinless fermions allow only 0 or 1 in each single-particle state.